Medical Tests and Investigations

1. Imaging / Radiology

- X-ray (Radiography)
- Computed Tomography Scan
- Magnetic Resonance Imaging
- Ultrasound Scan (Ultrasonography)
- Doppler Ultrasound
- Mammography (Breast X-ray Imaging)
- Fluoroscopy (Real-time X-ray Imaging)
- Positron Emission Tomography Scan
- Nuclear Medicine Scan
- Bone Densitometry (Dual-Energy X-ray Absorptiometry – DEXA)
- Angiography (Imaging of Blood Vessels)
- Barium Studies (Contrast X-ray of Gastrointestinal Tract)
- Intravenous Urography / Intravenous Pyelography

2. Cardiac / Cardiovascular

- Electrocardiogram (ECG)
- Echocardiography
- Stress Test / Exercise ECG
- Holter Monitoring
- Cardiac Catheterization
- Coronary Angiography
- Cardiac Biomarkers: Troponin, Creatine Kinase-MB, B-type Natriuretic Peptide
- Ambulatory Blood Pressure Monitoring

3. Hematology

- Complete Blood Count
- Peripheral Blood Smear
- Hemoglobin Electrophoresis
- Erythrocyte Sedimentation Rate
- Reticulocyte Count
- Platelet Count
- Coagulation Profile (PT, aPTT, INR)
- Bone Marrow Aspiration
- Bone Marrow Biopsy
- Platelet Function Assay

- D-Dimer

4. Clinical Chemistry

- Blood Glucose (Fasting, Random, Post-Prandial)
- Glycated Hemoglobin (HbA1c)
- Blood Urea
- Serum Creatinine
- Serum Electrolytes
- Liver Function Tests (ALT, AST, ALP, Bilirubin)
- Serum Albumin and Total Protein
- Albumin/Globulin Ratio
- Lipid Profile
- Urinalysis
- Amylase and Lipase
- Serum Osmolality
- Minerals: Calcium, Phosphate, Magnesium
- Iron Studies: Iron, TIBC, Ferritin

5. Microbiology

- Blood Culture
- Urine Culture
- Sputum Culture
- Stool Culture
- Wound/Swab Culture
- Cerebrospinal Fluid Culture
- Throat Swab Culture
- Antibiotic Sensitivity Testing
- Polymerase Chain Reaction Test
- Rapid Antigen Tests
- Viral Culture
- Fungal Culture
- Potassium Hydroxide Mount
- Stool Microscopy for Parasites
- Malaria Blood Smear
- Tuberculosis GeneXpert / AFB Smear

6. Immunology / Serology

- HIV Test
- Hepatitis Serology

- Rheumatoid Factor
- Antinuclear Antibody Test
- C-Reactive Protein
- Anti-Streptolysin O Titer
- Complement Levels
- Allergy Panels
- Autoimmune Panels

7. Endocrinology

- Thyroid Function Tests (TSH, T3, T4)
- Cortisol and ACTH
- Sex Hormones (FSH, LH, Estradiol, Progesterone, Testosterone)
- Prolactin
- Insulin Assay
- HbA1c

8. Toxicology / Drug Monitoring

- Blood Alcohol Test
- Toxicology Screen
- Therapeutic Drug Monitoring
- Poison Panels

9. Respiratory / Pulmonary

- Spirometry
- Peak Flow Test
- Arterial Blood Gas
- Pulse Oximetry
- Lung Function Tests
- Sleep Study (Polysomnography)

10. Neurology

- Electroencephalography (EEG)
- Nerve Conduction Study
- Electromyography

11. Gynecology & Maternity

- Pregnancy Test (Serum/Urine)

- Beta-hCG Quantitative Test
- Pap Smear
- Human Papillomavirus Test
- Vaginal/Cervical Swab Culture
- Hormonal Panels
- Obstetric/Pelvic Ultrasound
- Fetal Monitoring (CTG)
- Amniocentesis
- Quadruple Screen
- Group B Streptococcus Screening

12. Pediatrics

- Neonatal Screening Panel
- Heel-Prick Metabolic Screen
- Growth Hormone Testing
- Pediatric Genetic Tests

13. Surgery / Pre-Operative

- Type and Crossmatch
- Pre-operative Coagulation Profile
- Pre-operative Electrolytes and Renal Function
- Pre-operative Chest X-ray
- Pre-operative ECG
- Pre-operative Abdominal Ultrasound

14. Gastroenterology

- Endoscopy
- Colonoscopy
- Upper Gastrointestinal Study
- Liver Elastography
- Helicobacter pylori Testing

15. Urology

- Cystoscopy
- Renal Ultrasound
- Uroflowmetry
- Prostate Specific Antigen (PSA)

16. Orthopedics

- Bone Density Scan (DEXA)
- Bone Scan
- Joint X-ray
- MRI of Joint
- CT of Bone

17. Oncology / Tumor Markers

- Tumor Markers (AFP, CEA, CA-125, CA15-3, PSA)
- Biopsy (Tissue/Cytology)
- Flow Cytometry
- Genetic Tumor Panels

18. Pathology

- Histopathology
- Cytology
- Fine Needle Aspiration
- Tissue Biopsy
- Frozen Section
- Immunohistochemistry

19. Other / Miscellaneous

- Bone Marrow Examination
- Synovial Fluid Analysis
- Cerebrospinal Fluid Analysis
- Sweat Chloride Test
- Genetic Sequencing / Biochemical Genetic Tests

❖ AFTER THIS FOLLOW THE INSTRUCTION

Hospital Investigation Management System

Theoretical Proposal

1. Introduction

Hospitals perform a large number of diagnostic investigations such as laboratory tests, imaging procedures, and specialized examinations. However, not every hospital provides all possible investigations due to differences in equipment, laboratory capacity, staff availability, and operational policies.

To manage this efficiently, a Hospital Investigation Management System is required. This system maintains a complete master list of all medical investigations while allowing the hospital manager to activate only the investigations available in the hospital.

The system provides an easy interface where investigations can be searched, filtered by category, and activated or deactivated using checkboxes with visual indicators.

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2. System Objectives

The main objectives of the system are:

To maintain a centralized master list of all medical investigations.

To allow hospital administrators or managers to control which investigations are available in their hospital.

To improve clinical workflow for doctors when ordering tests.

To ensure patients only see investigations that are actually available.

To provide easy search and category-based filtering of investigations.

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3. System Structure

The system is designed with three main components:

1. Master Investigation Database

This database contains all possible medical investigations, categorized by department or medical specialty.

Examples of categories include:

Imaging / Radiology

Cardiology

Hematology

Clinical Chemistry

Microbiology

Immunology

Endocrinology

Respiratory Medicine

Neurology

Gynecology

Pediatrics

Surgery

Gastroenterology

Urology

Orthopedics

Oncology

Pathology

-

This master list ensures that the system always contains the complete range of investigations.

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2. Hospital Investigation Availability

Each hospital can select which investigations are available.

The hospital manager can:

- Activate investigations that the hospital provides

Deactivate investigations that are not available

The system stores this status as:

Active (Available)

Inactive (Not Available)

-

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4. Investigation Management Interface

The system includes a management dashboard where the hospital manager controls investigation availability.

Features of the interface include:

Search Function

Managers can search investigations by name.

Example:

Search: Blood

Results:

Blood Glucose

Blood Culture

Complete Blood Count

-

Category Filter

Investigations can be filtered by medical category using a dropdown menu.

Example categories:

Imaging / Radiology

Cardiac / Cardiovascular

Hematology

Microbiology

Clinical Chemistry

Neurology

Oncology

-

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5. Investigation Status Indicators

Each investigation has a visual status indicator.

Active Investigation

Checkbox: Checked

Color: Green

Status: AvailableExample:

☑ X-ray — □ Available

Inactive Investigation

Checkbox: Unchecked

Color: Red

Status: Not Available

Example:

☐ CT Scan — ☐ Not Available

---

6. Investigation Selection Process

The hospital manager performs the following steps:

Opens the Investigation Management Page

Searches or filters investigations

Activates investigations by checking the box

Deactivates investigations by unchecking the box

Saves the changes

1.

This allows the hospital to quickly configure its diagnostic services.

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7. Investigation Ordering by Doctors

When doctors order investigations for patients:

The system displays only active investigations.

Inactive investigations are hidden.

Available investigations:

✓ X-ray
✓ MRI
✓ ECG
✓ Blood Glucose

Unavailable investigations are not shown in the ordering interface.

---

8. Advantages of the System

The system provides several benefits:

1. Accurate Investigation Availability

Doctors will not request tests that are unavailable.

2. Efficient Hospital Management

Managers can easily control investigation availability.

3. Faster Clinical Workflow

Doctors can quickly find available investigations.

4. Flexible System Configuration

Hospitals can adjust services based on equipment or staffing.

5. Improved Patient Service

Patients receive only tests that the hospital can perform.

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9. Future Enhancements

The system can be expanded with additional features such as:

Investigation pricing management

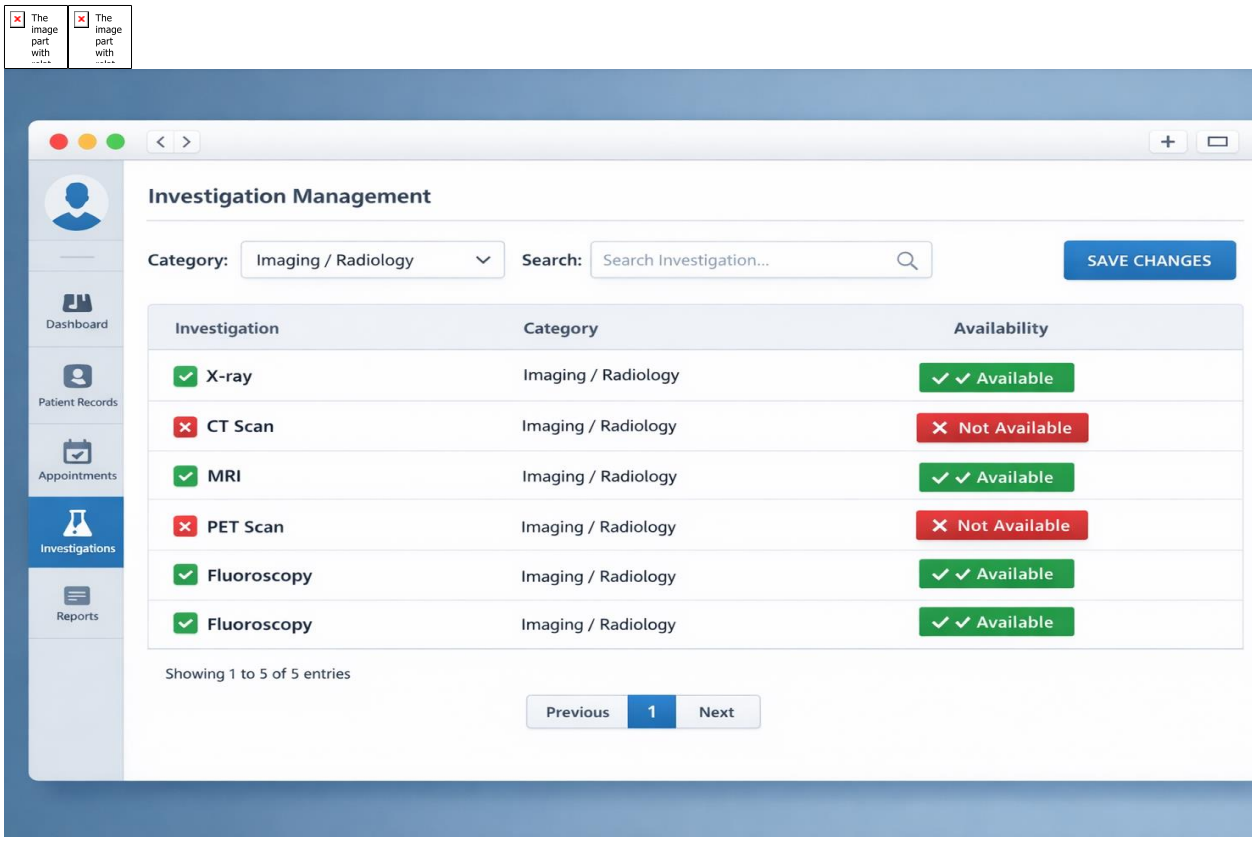
Equipment status monitoring

Department-based investigation assignment

Laboratory result integration

Investigation scheduling

Electronic medical record integration



Hospital Investigation Workflow in Your System

1. Patient Registration

First, the patient must exist in the system.

Each patient has a Health ID.

Example:

Patient Name	Health ID	Age
Ahmed Ali	HLT-000234	45
Sara Mohamed	HLT-000235	30

This Health ID connects all records:

-

Doctor visit

-
-

Investigations

-
-

Lab results

-
-

Diagnosis

-
-

Treatment

-

---

2. Doctor Opens Patient Record

Doctor searches patient:

```
Search Patient: Ahmed Ali
Health ID: HLT-000234
```

Doctor dashboard shows:

```
Patient Information
--------------------------------
Name: Ahmed Ali
Health ID: HLT-000234
Age: 45
Visit Date: 12/05/2026
--------------------------------
```

```
[ Diagnosis ]
[ Investigations ]
[ Prescription ]
```

Doctor clicks Investigations.

---

3. Doctor Selects Investigation

System shows only ACTIVE investigations.

Example:

```
Investigation Order
--------------------------------

Category: [ Laboratory ▼ ]
Search: [_________]

Available Investigations

☑ Complete Blood Count
☑ Blood Glucose
☑ Lipid Profile
☑ Liver Function Test
☑ Urine Analysis

--------------------------------

[ Submit Investigation Order ]
```

Doctor selects needed tests.

Example selected:

☑ Complete Blood Count
☑ Blood Glucose
☑ Liver Function Test

Then doctor clicks:

Submit Order

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4. Investigation Order Sent to Department

After submission, the system automatically sends the order to the correct department.

Example routing:

Investigation	Department
CBC	Laboratory
Blood Glucose	Laboratory
MRI	Radiology
ECG	Cardiology

Order example:

New Investigation Request

Patient: Ahmed Ali
Health ID: HLT-000234

Requested Tests

- Complete Blood Count
- Blood Glucose
- Liver Function Test

```
Status: Pending
```

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5. Laboratory / Radiology Technician Portal

Lab technician logs in.

Portal shows pending investigation requests.

Example:

```
Laboratory Dashboard

Pending Tests
--------------------------------

Patient: Ahmed Ali
Health ID: HLT-000234

Tests Requested

• Complete Blood Count
• Blood Glucose
• Liver Function Test

Status: Pending

[ Start Test ]
```

After performing tests, technician enters results.

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6. Technician Submits Result

Technician opens patient:

```
Patient: Ahmed Ali
Health ID: HLT-000234
```

Then enters results:

```
Complete Blood Count

WBC: 6.5
RBC: 4.8
Hemoglobin: 13.5

Blood Glucose

Result: 120 mg/dL

Liver Function Test

ALT: 32
AST: 28
```

Technician clicks:

```
Submit Result
```

System status becomes:

□ Result Available

---

7. Doctor Receives Result

Doctor opens patient again.

Now system shows:

```
Investigation Status

Complete Blood Count       Result Available
```

```
Blood Glucose              Result Available
Liver Function Test        Result Available
```

Doctor clicks View Result.

---

8. Doctor Reviews Result

Example screen:

```
Patient: Ahmed Ali
Health ID: HLT-000234

Lab Results
--------------------------------

Complete Blood Count
WBC: 6.5
RBC: 4.8
Hemoglobin: 13.5

Blood Glucose
Result: 120 mg/dL

Liver Function
ALT: 32
AST: 28
--------------------------------
```

---

9. Doctor Makes Diagnosis

Doctor writes diagnosis.

Example:

Diagnosis

Mild Hyperglycemia

Plan

- Lifestyle modification
- Follow up in 1 month
- Repeat blood glucose

---

10. Final Patient Record

All information stored in patient history:

Patient History

Visit Date: 12/05/2026

Doctor: Dr. John

Investigations

✔CBC
✔Blood Glucose
✔Liver Function Test

Results: Available

Diagnosis: Mild Hyperglycemia
Treatment: Medication + Diet Control

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11. Investigation Status Colors

System color logic:

Status	Color
Ordered	Yellow
In Progress	Blue
Result Ready	Green
Cancelled	Red

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12. System Modules Involved

Your HIS system modules:

Patient Registration
Doctor Consultation
Investigation Ordering
Laboratory Management
Radiology Management
Result Reporting
Diagnosis & Treatment